

# VISITOR APPROVAL ENTRY SYSTEM

*A full-stack visitor registration and approval platform — React, Node/Express & MySQL*

Database Systems Engineering and Distributed Backend Development (25CS1302E)

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# Abstract

- 1 What it is**

A digital gate-security app for apartment/office visitor entry, with separate Guard and Tenant portals.
- 2 Problem solved**

Replaces manual visitor registers and unverified walk-ins with a logged, approval-based entry process.
- 3 Core flow**

Guard registers a visitor → request sent to the tenant → tenant approves/denies → guard sees the result.
- 4 Tech used**

React (Vite) frontend, Node.js/Express REST API, MySQL database with bcrypt-hashed passwords.
- 5 Benefit**

Faster, traceable visitor handling with a permanent visit log and role-based access.

# Problem Statement

## Manual registers

Paper logs are slow, illegible, and easy to falsify or lose.

## No approval step

Walk-in visitors often enter before the resident/host confirms them.

## Poor traceability

No searchable history of who visited, when, or why.

## No absence handling

Manual systems can't verify a visitor when the host is away.

*A digital, role-based approval workflow is needed to close these gaps.*

| Existing Approach            | Method                          | Limitation                                     | VAES Improvement                                               |
|------------------------------|---------------------------------|------------------------------------------------|----------------------------------------------------------------|
| Paper visitor register       | Handwritten log at gate         | Slow, illegible, no approval step              | Digital form + instant tenant approval/denial                  |
| Gate phone call to resident  | Guard calls resident to confirm | No record kept; depends on call being answered | Absence-password fallback + stored visit log                   |
| Temporary approval generator | No storage                      | History is not recorded                        | Web-based, role-separated Guard/Tenant dashboards with history |

*Research gap: most classroom/reference visitor-management approaches log entries but lack a real-time, host-side approve/deny workflow with role separation the gap VAES targets.*

Note: comparisons reflect common manual/legacy approaches; no specific published papers were provided in the project files.

# Objectives

- 1 Register visitor details digitally (name, phone, purpose, host block/flat)
- 2 Route each visit request to the tenant for approval or rejection
- 3 Support pre-approval of expected guests, so entry is instant
- 4 Allow tenants to schedule expected visits in advance
- 5 Maintain a permanent, searchable visit log for every entry
- 6 Provide secure, role-based login for guards and tenants

# Proposed System

VAES is a two-portal web app (Guard / Tenant) backed by a REST API and MySQL database.

1

## Authentication

Login & signup for tenants/guards; bcrypt password hashing

2

## Security Module

Guard dashboard: register visitors, view sent requests

3

## Tenant Module

Tenant dashboard: approvals, pre-approvals, schedules

4

## Visitor Management

Visitor entry form; via Pre-Approved / Schedule / Direct

5

## Approval Management

Tenant approves/denies pending visitor requests

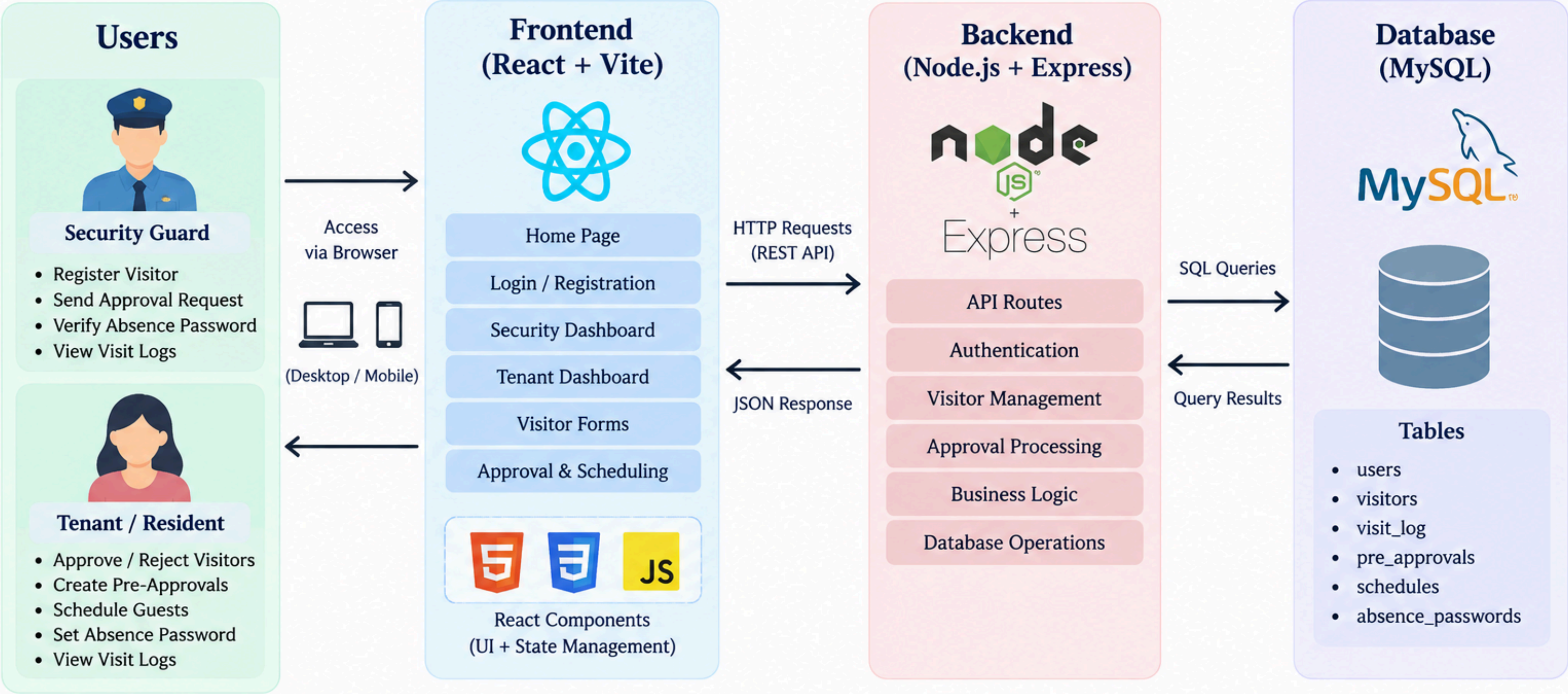
6

## Visit Logs

Append-only historical log of every visitor record

# System Architecture

## Visitor Approval Entry System (VAES)



### Technology Stack



React  
(Vite)



Node.js  
(Express)



MySQL  
(Database)



HTML5



CSS3



JavaScript



Visual Studio Code  
(Development)

A Secure and Efficient Digital Solution for Visitor Management

# FRONTEND DESIGN

## Home Page

- Introduces the Visitor Approval Entry System (VAES)
- Provides navigation to the system
- Clean and user-friendly interface

## Login / Registration

- Tenant registration
- Security registration
- Secure authentication

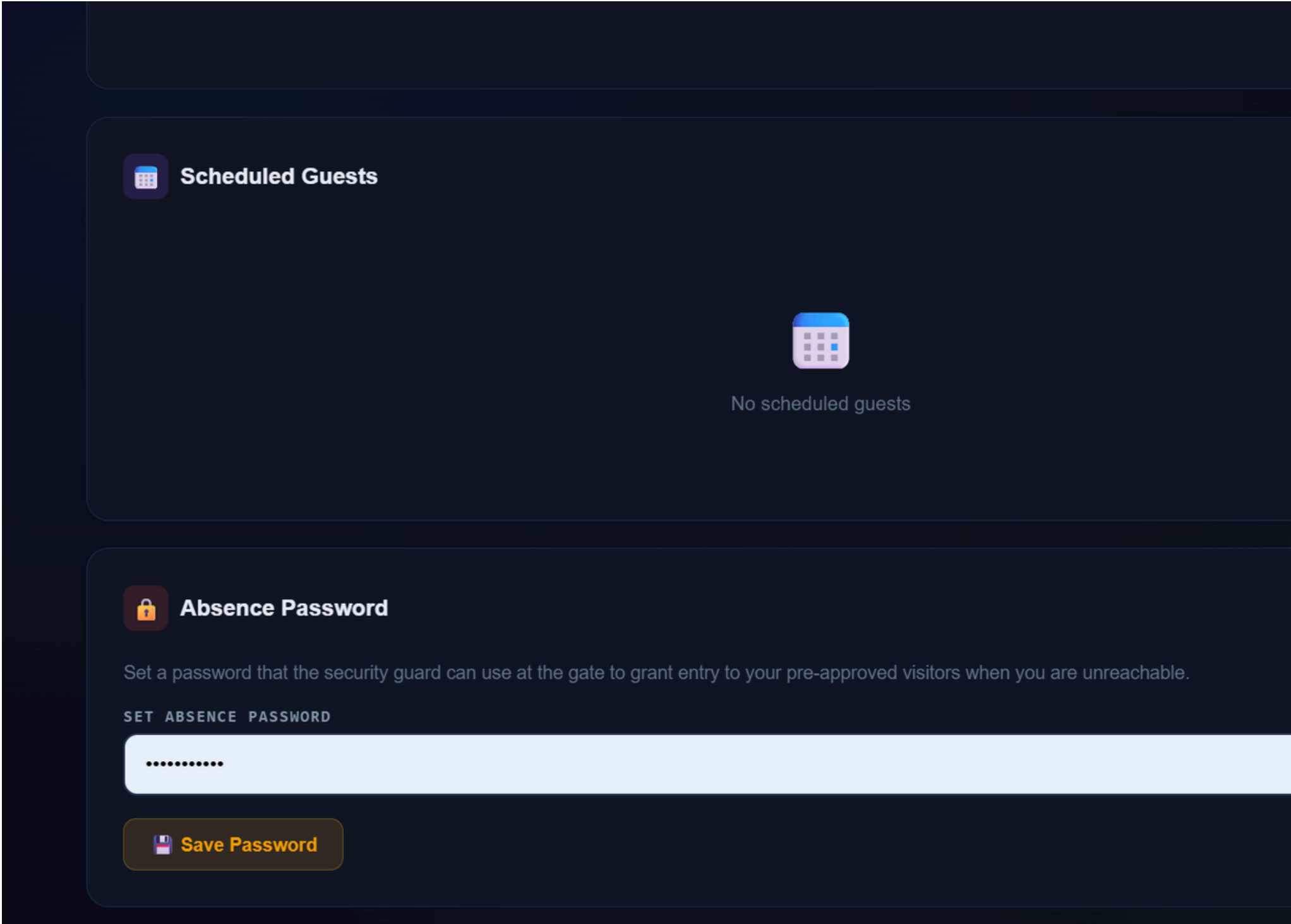
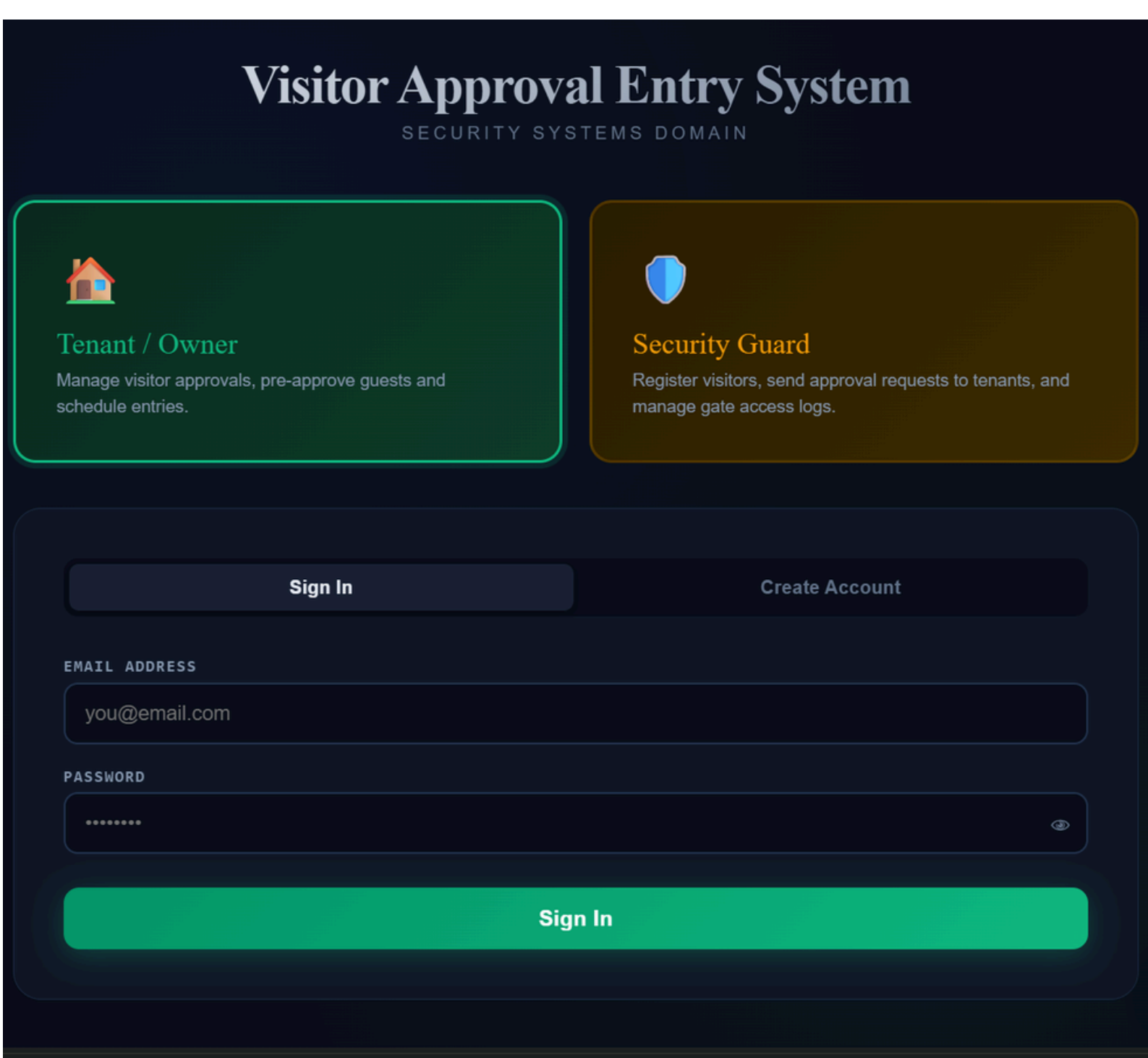
## forms

- **Visitor registration form**
- **Visitor details: name, contact, purpose, host**
- **Pre-approval and guest scheduling forms**
- **Input validation**

## Dashboard

- Security Dashboard: visitor registration, approval requests and visit logs
- Tenant Dashboard: approve/reject requests, pre-approvals, scheduling and visit history

# FRONTEND Implementation Screenshots



# Database Design

MySQL database "vaes" — 6 tables, linked mainly through email / schedule\_id

## users

- PK id
- role (tenant/guard)
- email (UNIQUE)
- password (bcrypt)
- block, flat

## absence\_passwords

- PK/FK user\_email
- password (bcrypt)
- → users.email

## pre\_approvals

- PK id
- name, phone, relation
- block, flat

## schedules

- PK id
- name, phone, date/time
- block, flat

## visitors

- PK id
- status  
(pending/approved/denied)
- FK schedule\_id
- block, flat, guard\_id

## visit\_log

- PK id (append-only)
- same fields as visitors
- permanent history

# Database Tables/Schema

|             |               |            |              |                      |            |          |       |      |                     |  |  |                    |  |
|-------------|---------------|------------|--------------|----------------------|------------|----------|-------|------|---------------------|--|--|--------------------|--|
| Result Grid |               |            | Filter Rows: | <input type="text"/> | Edit:      |          |       |      | Export/Import:      |  |  | Wrap Cell Content: |  |
|             | id            | name       | phone        | relation             | until      | schedule | block | flat | created_at          |  |  |                    |  |
| ▶           | 9000000000001 | Ravi Kumar | 9876543210   | Brother              | 2026-12-31 |          | A     | 101  | 2026-09-11 14:48:40 |  |  |                    |  |
|             | 9000000000002 | Anjali Rao | 9123456780   | Friend               | 2026-10-15 |          | A     | 101  | 2026-09-11 14:48:40 |  |  |                    |  |
| ✱           | NULL          | NULL       | NULL         | NULL                 | NULL       | NULL     | NULL  | NULL | NULL                |  |  |                    |  |

|             |               |                  |              |                      |       |                  |       |      |                     |  |  |                    |  |
|-------------|---------------|------------------|--------------|----------------------|-------|------------------|-------|------|---------------------|--|--|--------------------|--|
| Result Grid |               |                  | Filter Rows: | <input type="text"/> | Edit: |                  |       |      | Export/Import:      |  |  | Wrap Cell Content: |  |
|             | id            | name             | phone        | date                 | time  | purpose          | block | flat | created_at          |  |  |                    |  |
| ▶           | 9000000000101 | Amazon Delivery  | 9000011111   | 2026-09-15           | 14:00 | Package delivery | A     | 101  | 2026-09-11 14:48:40 |  |  |                    |  |
|             | 9000000000102 | Plumber - Suresh | 9000022222   | 2026-09-18           | 10:30 | Leak repair      | B     | 402  | 2026-09-11 14:48:40 |  |  |                    |  |
| ✱           | NULL          | NULL             | NULL         | NULL                 | NULL  | NULL             | NULL  | NULL | NULL                |  |  |                    |  |

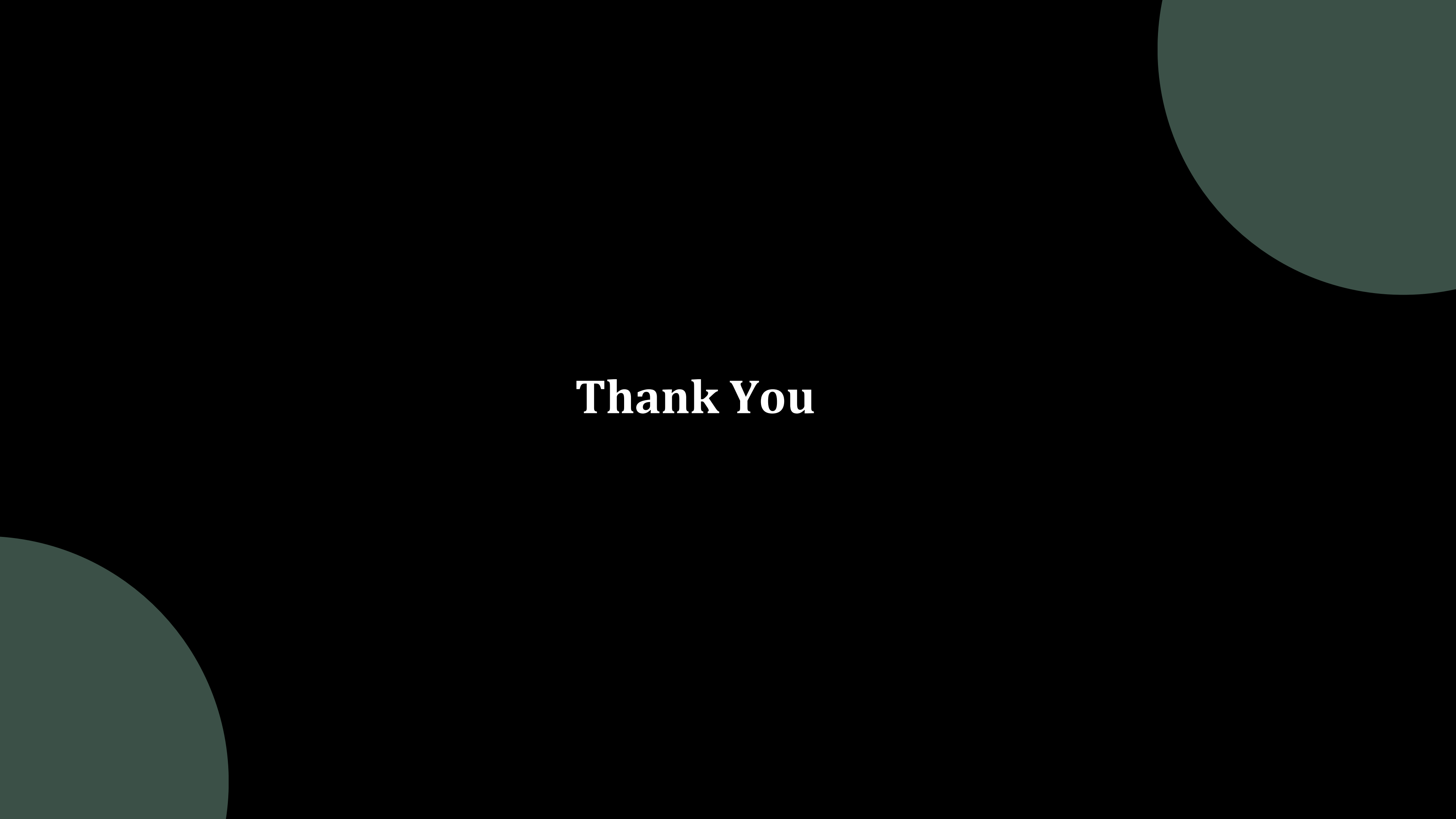
|             |               |                        |              |                      |       |      |          |                   |                       |                 |             |                    |  |
|-------------|---------------|------------------------|--------------|----------------------|-------|------|----------|-------------------|-----------------------|-----------------|-------------|--------------------|--|
| Result Grid |               |                        | Filter Rows: | <input type="text"/> | Edit: |      |          |                   | Export/Import:        |                 |             | Wrap Cell Content: |  |
|             | id            | name                   | phone        | purpose              | block | flat | status   | via               | time                  | guard_id        | schedule_id |                    |  |
| ▶           | 9000000000201 | Deepak Sharma          | 9988776655   | Meeting friend       | A     | 101  | approved | Host Approved     | 11/9/2026, 2:48:40 pm | ramya@gmail.com | NULL        |                    |  |
|             | 9000000000202 | Unknown Visitor        | 9111222333   | Sales pitch          | C     | 202  | denied   | Host Denied       | 11/9/2026, 2:48:40 pm | ramya@gmail.com | NULL        |                    |  |
|             | 9000000000203 | Courier - Swiggy Genie | 9333444555   | Package drop         | B     | 402  | pending  | Awaiting Approval | 11/9/2026, 2:48:40 pm | ramya@gmail.com | NULL        |                    |  |
| ✱           | NULL          | NULL                   | NULL         | NULL                 | NULL  | NULL | NULL     | NULL              | NULL                  | NULL            | NULL        |                    |  |

# Challenges / Issues Faced

- **React conversion –**  
Converting the existing frontend into reusable React components.
- **Frontend–Backend Integration –**  
Connecting React with the Node.js/Express REST API.
- **MySQL Connectivity –**  
Configuring database connection and environment variables.
- **Database Setup –**  
Creating tables, relationships and inserting sample records.
- **State Management –**  
Managing shared user, visitor and approval data using React Context.
- **Data Validation –**  
Handling form inputs and preventing invalid visitor information.
- **Debugging –**  
Resolving errors during frontend, backend and database integration.

# Results / Benefits

- |          |                            |                                                                             |
|----------|----------------------------|-----------------------------------------------------------------------------|
| <b>1</b> | <b>Faster processing</b>   | Digital form + instant pre-approval removes manual write-ups at the gate    |
| <b>2</b> | <b>Centralized records</b> | All visitor, approval, and log data stored in one MySQL database            |
| <b>3</b> | <b>Better security</b>     | Every entry needs tenant approval or a valid pre-approval/absence password  |
| <b>4</b> | <b>Easier tracking</b>     | Append-only visit_log gives a full, searchable visitor history              |
| <b>5</b> | <b>Modular React UI</b>    | Reusable components (Modal, StatCard, Toast) keep the codebase maintainable |



**Thank You**